

Practice Sheet for DSA-1 Lab

Trimester: Spring 26

Problem 1

Given the head of a linked list, print the size of the linked list.

Example Input/Output

Input: 10 → 20 → 30 → 40

Output: 4

Problem 2

Given the head of a linked list, find the middle node of the linked list.

Example Input/Output

Odd-sized list

Input: 1 → 2 → 3 → 4 → 5

Output: 3

Even-sized list

Input: 1 → 2 → 3 → 4 → 5 → 6

Output: 4

Problem 3

Given the head of a linked list and a number, print how many times the given number occurs in the linked list.

Example Input/Output

Input list: 4 → 7 → 4 → 9 → 4

Given number: 4

Output: 3

Problem 4

Create a linked list to store these three characters: 'a' → 'b' → 'c'.

Example Input/Output**Input:** a, b, c**Output:** a $\rightarrow$ b $\rightarrow$ c**Problem 5**

Search a linked list for a given number. If the number is found, print **Yes**; otherwise, print **No**.

Example Input/Output**Input list:** 8 $\rightarrow$ 3 $\rightarrow$ 12 $\rightarrow$ 5**Given number:** 12**Output:** Yes**Problem 6**

Given the head of a linked list, insert a new node before the head.

Example Input/Output**Input list:** 20 $\rightarrow$ 30 $\rightarrow$ 40**New value:** 10**Output:** 10 $\rightarrow$ 20 $\rightarrow$ 30 $\rightarrow$ 40**Problem 7**

Given the head of a linked list, insert a new node at the end of the linked list.

Example Input/Output**Input list:** 11 $\rightarrow$ 22 $\rightarrow$ 33**New value:** 44**Output:** 11 $\rightarrow$ 22 $\rightarrow$ 33 $\rightarrow$ 44**Problem 8**

Given the head of a linked list, insert a new node after a given node. The value of the given node will be provided.

Example Input/Output**Input list:** $5 \rightarrow 10 \rightarrow 20 \rightarrow 30$ **Insert after value:** 10**New value:** 15**Output:** $5 \rightarrow 10 \rightarrow 15 \rightarrow 20 \rightarrow 30$ **Problem 9**

Given the head of a linked list, delete the head node.

Example Input/Output**Input list:** $9 \rightarrow 18 \rightarrow 27 \rightarrow 36$ **Output:** $18 \rightarrow 27 \rightarrow 36$ **Problem 10**

Given the head of a linked list, delete the last node of the linked list.

Example Input/Output**Input list:** $9 \rightarrow 18 \rightarrow 27 \rightarrow 36$ **Output:** $9 \rightarrow 18 \rightarrow 27$ **Problem 11**

Given the head of a linked list, delete the node after a given node. The value of the given node will be provided.

Example Input/Output**Input list:** $2 \rightarrow 4 \rightarrow 6 \rightarrow 8 \rightarrow 10$ **Delete node after value:** 4**Output:** $2 \rightarrow 4 \rightarrow 8 \rightarrow 10$ **Doubly Linked List Problems****Problem 12**

Create a doubly linked list to store the values: 11, 22, 33.

Example Input:

11 22 33

Example Output:

11 ↔ 22 ↔ 33

Problem 13

Insert a new node at the head of a doubly linked list.

Example Input:

List: 11 ↔ 22 ↔ 33

New Value: 5

Example Output:

5 ↔ 11 ↔ 22 ↔ 33

Problem 14

Delete the head node of a doubly linked list.

Example Input:

List: 11 ↔ 22 ↔ 33

Example Output:

22 ↔ 33